Homework 9

1) Given

$$P(y_i=1) = p_i = \frac{e^{x_i\beta}}{1+e^{x_i\beta}}, \quad y_i \in \{0,1\}, \quad i = 1,\dots,n$$

This is a logistic regression model with one parameter β.

i) To find, Log-likelihood function $L(\beta)$

For Bernoulli, $P(y_i) = p_i^{y_i}(1-p_i)^{1-y_i}$

Substitute p_i from given,

$$P(y_i) = \left(\frac{e^{x_i\beta}}{1+e^{x_i\beta}}\right)^{y_i}\left(\frac{1}{1+e^{x_i\beta}}\right)^{1-y_i}$$

$$= \frac{e^{x_i\beta y_i}}{(1+e^{x_i\beta})^{y_i}(1+e^{x_i\beta})^{1-y_i}}$$

$$\Rightarrow P(y_i) = \frac{e^{x_i\beta y_i}}{(1+e^{x_i\beta})}$$

Notes

Likelihood: $L(\beta) = \prod_{i=1}^{n} \frac{e^{x_i \beta y_i}}{1+e^{x_i\beta}}$

Log-likelihood: $\log L(\beta) = \sum_{i=1}^{n} \left[x_i \beta y_i - \log(1+e^{x_i\beta})\right]$

ii) To find First & Second Derivatives.

First derivative: $\frac{dL}{d\beta} = \frac{d}{d\beta} \log L(\beta)$

$$= \frac{d}{d\beta} \sum_{i=1}^{n} \left[x_i \beta y_i - \log(1+e^{x_i\beta})\right]$$

$$= \sum_{i=1}^{n} \left[x_i y_i - \frac{x_i e^{x_i\beta}}{1+e^{x_i\beta}}\right]$$

We know that $p_i = \frac{e^{x_i\beta}}{1+e^{x_i\beta}}$

$$\Rightarrow \frac{dL}{d\beta} = \sum_{i=1}^{n} x_i (y_i - p_i)$$

Second derivative: $\frac{d^2L}{d\beta^2} = \sum_{i=1}^{n} \left[-x_i^2 \frac{e^{x_i\beta}}{(1+e^{x_i\beta})^2}\right]$

Using $p_i(1-p_i)$

08.00

$$\Rightarrow \frac{d^2 L}{d\beta^2} = -\sum_{i=1}^{n} x_i^2 p_i(1-p_i)$$

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iii) 11.00 Newton Raphson update:

$$\beta^{(t+1)} = \beta^{(t)} - \frac{L'(\beta)}{L''(\beta)}$$

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We know that,

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$$L'(\beta) = \sum_{i=1}^{n} x_i(y_i - p_i)$$

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$$L''(\beta) = -\sum_{i=1}^{n} x_i^2 p_i(1-p_i)$$

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17.00

$$\Rightarrow \beta^{(t+1)} = \beta^{(t)} + \frac{\sum_{i=1}^{n} x_i(y_i - p_i)}{\sum_{i=1}^{n} x_i^2 p_i(1-p_i)}$$

18.00

Notes

where

$$p_i = \frac{e^{x_i\beta(t)}}{1+e^{x_i\beta(t)}}$$

iv) To show,

$$E(\hat{\beta}) = \beta, \quad Var(\hat{\beta}) = \frac{1}{\sum_{i=1}^{n} x_i^2 \hat{p}_i (1-\hat{p}_i)}$$

For maximum likelihood estimators:

$$\hat{\beta} \approx N\left(\beta, \frac{1}{I(\beta)}\right)$$ where $I(\beta)$ is the Fisher information

Recall log likelihood, $\ell(\beta) = \sum_{i=1}^{n} [x_i \beta y_i - \log(1+e^{x_i\beta})]$

By Fisher Information definition

$$I(\beta) = -E[\ell''(\beta)]$$

We know ~~xx~~ $\ell''(\beta)$ from previous derivations.
Substituting, we get,

$$I(\beta) = -E\left[-\sum_{i=1}^{n} x_i^2 p_i (1-p_i)\right]$$

$$I(\beta) = \sum_{i=1}^{n} x_i^2 p_i (1-p_i)$$

p_i depends on x_i, β, not on random y_i

So, expectation doesn't change it.

Now, $Var(\hat{\beta}) = \dfrac{1}{I(\beta)}$

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09.00 $\Rightarrow Var(\hat{\beta}) = \dfrac{1}{\sum_{i=1}^{n} x_i^2 p_i(1-p_i)}$

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Since β is unknown, we replace with MLE $\hat{\beta}$

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$$\hat{p}_i = \frac{e^{x_i\hat{\beta}}}{1+e^{x_i\hat{\beta}}}$$

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13.00 $\Rightarrow Var(\hat{\beta}) = \dfrac{1}{\sum_{i=1}^{n} x_i^2 \hat{p}_i(1-\hat{p}_i)}$

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15.00 From MLE theory, $\hat{\beta} \xrightarrow{\text{asymptotically}} N(\beta, \ldots)$

16.00 $\Rightarrow E(\hat{\beta}) \approx \beta$

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v) Hypothesis test: $H_0 : \beta = 0$

Notes

We use a Wald test.

Test Statistic: $Z = \dfrac{\hat{\beta}}{\sqrt{Var(\hat{\beta})}}$

$$\Rightarrow Z = \hat{\beta} \cdot \sqrt{\sum_{i=1}^{n} x_i^2 \hat{p}_i (1-\hat{p}_i)}$$

Decision Rule, Under H_0, $Z \sim N(0,1)$

Reject if: $|Z| > z_{\alpha/2}$

If rejected $\rightarrow$ x significantly affects y.

If not $\rightarrow$ no significant effect.

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2) We now take a Bayesian approach:

Likelihood $\rightarrow$ Same as problem 1.

Prior $\rightarrow \beta \sim N(0, \sigma^2)$

i) Posterior Density (up to proportionality)

From Problem 1, $L(\beta) = \prod_{i=1}^{n} \dfrac{e^{x_i \beta y_i}}{1+e^{x_i \beta}}$

NOVEMBER 2025					
Sun	30	2	9	16	23
Mon		3	10	17	24
Tue		4	11	18	25
Wed		5	12	19	26
Thu		6	13	20	27
Fri		7	14	21	28
Sat	1	8	15	22	29

DECEMBER 2025				
Sun		7	14	21
Mon	1	8	15	22
Tue	2	9	16	23
Wed	3	10	17	24
Thu	4	11	18	25
Fri	5	12	19	26
Sat	6	13	20	27

08.00 Normal prior, $\pi(\beta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta^2}{2\sigma^2}\right)$

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Apply Bayes' Theorem,

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$$\pi(\beta \mid data) \propto L(\beta)\,\pi(\beta)$$

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Multiply Likelihood & Prior.

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$$\pi(\beta \mid data) \propto \left[\prod_{i=1}^{n} \frac{e^{x_i \beta y_i}}{1+e^{x_i \beta}}\right] \cdot \exp\left(-\frac{\beta^2}{2\sigma^2}\right)$$

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14.00 Combine Exponent Terms,

$$\prod e^{x_i \beta y_i} = e^{\sum x_i \beta y_i}$$

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$$\Rightarrow \pi(\beta \mid data) \propto \exp\left(\sum_{i=1}^{n} x_i \beta y_i\right) \cdot \exp\left(-\frac{\beta^2}{2\sigma^2}\right) \cdot \prod_{i=1}^{n} \frac{1}{1+e^{x_i \beta}}$$

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$$\Rightarrow \pi(\beta \mid data) \propto \exp\left(\sum_{i=1}^{n} x_i \beta y_i - \frac{\beta^2}{2\sigma^2}\right) \prod_{i=1}^{n} \frac{1}{1+e^{x_i \beta}}$$

Notes

JANUARY 2026					
Sun		4	11	18	25
Mon		5	12	19	26
Tue		6	13	20	27
Wed		7	14	21	28
Thu	1	8	15	22	29
Fri	2	9	16	23	30
Sat	3	10	17	24	31

FEBRUARY 2026				
Sun	1	8	15	22
Mon	2	9	16	23
Tue	3	10	17	24
Wed	4	11	18	25
Thu	5	12	19	26
Fri	6	13	20	27
Sat	7	14	21	28

ii) Show Log Posterior is Concave.

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$$\log \pi(\beta \mid data) = \sum_{i=1}^{n} x_i \beta y_i - \sum_{i=1}^{n} \log(1+e^{x_i\beta}) - \frac{\beta^2}{2\sigma^2} + C$$

09.00

10.00 First Derivative,

$$\frac{d}{d\beta} = \sum_{i=1}^{n} x_i (y_i - p_i) - \frac{\beta}{\sigma^2}$$

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12 Second Derivative,

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$$\frac{d^2}{d\beta^2} = -\sum_{i=1}^{n} x_i^2 p_i (1-p_i) - \frac{1}{\sigma^2}$$

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$\Rightarrow p_i(1-p_i) > 0$

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$\Rightarrow x_i^2 > 0$

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$\Rightarrow \frac{1}{\sigma^2} > 0$

So, $\frac{d^2}{d\beta^2} < 0$

11 Sunday

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$\Rightarrow$ Log posterior is concave //

Notes

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iii) Improper prior $\pi(\beta) \propto 1$

Posterior becomes, $\pi(\beta | data) \propto L(\beta)$

$\Rightarrow$ Posterior $\propto$ Likelihood

$\Rightarrow$ Maximizing posterior $\equiv$ Maximizing Likelihood

Therefore, the relationship between posterior & maximum likelihood estimator is that

$$\boxed{\text{Posterior mode} = \text{MLE } \hat{\beta}}$$

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iv) Log Bayes Factor:

We compare: $H_0 : \beta = 0$
$H_1 : \beta = 1$

By Bayes Factor definition,

$$B = \frac{L(\beta = 1)}{L(\beta = 0)}$$

Likelihood, $L(\beta) = \prod \frac{e^{x_i \beta y_i}}{1+e^{x_i \beta}}$

Take Log, $\log B = \log L(1) - \log L(0)$

Evaluate Each Term,

For $\beta = 1$: $\log L(1) = \sum x_i y_i - \sum \log(1+e^{x_i})$

For $\beta = 0$: $\log L(0) = \sum 0 - \sum \log(2) = -n \log 2$

$\Rightarrow \log B = \sum x_i y_i - \sum \log(1+e^{x_i}) + n \log 2$

Rearranged form:

$$\log B = \sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} \log(1+e^{x_i})$$

v) Approximate Distribution for $\log B$

We know that,

$$\log B = \sum_{i=1}^{n} \left[x_i y_i - \log(1+e^{x_i}) \right]$$

Under $H_0 : \beta = 0$

$$\Rightarrow p_i = \frac{1}{2}, \quad y_i \sim \text{Bernoulli}(1/2)$$

Mean, $E[x_i y_i] = x_i \cdot \frac{1}{2}$

$$\Rightarrow E[\log B] = \sum \left[x_i/2 - \log(1 + e^{x_i}) \right]$$

Variance, $Var(x_i y_i) = x_i^2 \cdot \frac{1}{4}$

$$\Rightarrow Var(\log B) = \sum \frac{x_i^2}{4}$$

Apply CLT, since sum of independent terms,

$$\log B \approx N\left(\sum \left[\frac{x_i}{2} - \log(1 + e^{x_i}) \right], \sum \frac{x_i^2}{4} \right)$$